



Digital Library Management Platform

Summary

A comprehensive web-based platform that modernizes library operations for students and faculty. Users can search the entire catalog with advanced filters, check real-time availability of physical and digital resources, access e-books and journals instantly, and place reservation requests with automated notifications.

The system streamlines traditional library processes by providing 24/7 access to resources from any location, eliminating manual searches and in-person visits. This centralized solution enhances accessibility, optimizes resource utilization, and creates an efficient ecosystem that supports the academic community's research and learning needs.

Definitions

Student

A registered college/university learner who uses the system to search, borrow, reserve, and access library resources for academic purposes with standard borrowing privileges.

Faculty Member

An instructor, professor, or academic staff member with enhanced library privileges including extended borrowing periods, higher checkout limits, and priority access to reserved materials for teaching and research purposes.

Librarian/Administrator

A library staff member responsible for managing the system, including adding/updating catalog entries, processing reservation requests, managing user accounts, tracking overdue items, and generating reports on library usage and inventory.

Book/Physical Resource

A tangible library item (books, journals, magazines, dissertations, etc.) with attributes including title, author, ISBN, publication details, category, location, and availability status that can be borrowed or reserved.

Digital Resource

Electronic materials such as e-books, online journals, research papers, databases, and multimedia content that can be accessed instantly without physical checkout, often with concurrent user limits.

Reservation/Hold Request

A user-initiated request to reserve a currently unavailable resource, queued in order of submission and automatically notified upon availability for pickup or access.

Transaction/Checkout Record

A record tracking the borrowing activity including user details, resource borrowed, checkout date, due date, return date, and any associated fines or renewal history.

Category/Classification

A taxonomic grouping system (such as Dewey Decimal or Library of Congress) used to organize resources by subject, discipline, or type for easier discovery and browsing.

Notification

Automated system alerts sent to users regarding reservation availability, due date reminders, overdue notices, and new resource additions matching their interests.

Fine/Penalty

A monetary charge applied to users for overdue returns, damaged materials, or lost items, tracked within the system and requiring clearance before further borrowing privileges are granted.

User Stories

Search / Browse Resources

As a student or faculty member

I want to search the library catalog using keywords, filters, and categories

So that I can quickly find relevant academic resources for my research

Acceptance Criteria

- User can search by title, author, ISBN, or keywords
- User can filter by category, availability, resource type, and publication year
- Search results display availability status in real-time
- User can view detailed information about each resource

Check Out Physical Resources

As a student or faculty member

I want to borrow available physical books and materials

So that I can use them for my studies and research

Acceptance Criteria

- User can view their current borrowing limit and active checkouts
- System displays due dates for all borrowed items
- User receives confirmation of successful checkout
- System prevents checkout if user has overdue items or unpaid fines
- Faculty members have extended checkout periods compared to students

Access Digital Resources

As a student or faculty member

I want to instantly access and download digital materials

So that I can review them without visiting the library

Acceptance Criteria

- User can view and download e-books, journals, and research papers
- System tracks concurrent usage limits for licensed content
- User can bookmark and save digital resources to their account
- Digital resources are accessible 24/7 from any device

Place Reservation Requests

As a student or faculty member

I want to reserve currently unavailable resources

So that I can be notified when they become available

Acceptance Criteria

- User can place holds on checked-out items
- System displays queue position for popular items
- User receives automated notification when resource is available
- Reservation expires after specified pickup period
- Faculty members receive priority in reservation queues

Manage Library Catalog

As a librarian/administrator

I want to add, update, and remove resources from the catalog

So that I can maintain accurate library inventory

Acceptance Criteria

- Librarian can add new resources with complete bibliographic details
- Librarian can update availability status and location
- Librarian can mark items as lost, damaged, or under repair

- System prevents deletion of resources with active checkouts
- Bulk import functionality available for large catalog updates

Process Checkouts and Returns

As a librarian/administrator

I want to manage checkout and return transactions

So that I can track resource circulation efficiently

Acceptance Criteria

- Librarian can scan or manually enter resource IDs for checkout
- System automatically calculates due dates based on user type
- Librarian can process returns and update availability instantly
- System calculates fines for overdue items automatically
- Librarian can renew items if no reservations exist

Monitor Overdue Items and Fines

As a librarian/administrator

I want to track overdue resources and manage fines

So that I can ensure timely returns and resource availability

Acceptance Criteria

- Dashboard displays all overdue items with user details
- System sends automated reminders before and after due dates
- Librarian can view and manage user fine balances
- Librarian can waive or adjust fines with proper authorization
- System generates overdue reports for administrative review

Generate Reports and Analytics

As a librarian/administrator

I want to generate reports on library usage and inventory

So that I can make informed decisions about resource acquisition

Acceptance Criteria

- System provides circulation statistics by category and time period
- Reports show most popular and least used resources
- Dashboard displays key metrics (total checkouts, reservations, overdue items)
- Librarian can export reports in multiple formats (PDF, Excel, CSV)
- Analytics track user engagement and resource utilization trends

Renew Borrowed Resources

As a student or faculty member

I want to renew my borrowed items before they're due

So that I can extend my access without returning and re-borrowing

Acceptance Criteria

- User can view all borrowed items with renewal eligibility status
- System allows renewal if no reservations exist for the item
- Users can renew items up to a maximum number of times (e.g., 2 renewals)
- System automatically updates due dates upon successful renewal
- User receives confirmation notification after renewal

View Borrowing History

As a student or faculty member

I want to view my complete borrowing and access history

So that I can track resources I've used for my research and citations

Acceptance Criteria

- User can view all previously borrowed physical resources
- User can view history of accessed digital resources
- History displays checkout dates, return dates, and resource details
- User can search and filter their history by date or resource type
- User can export their history for personal records

Receive Notifications and Reminders

As a student or faculty member

I want to receive timely notifications about my library activities

So that I can avoid overdue fines and stay informed about reservations

Acceptance Criteria

- User receives reminder 3 days before due date
- User receives notification when reserved item becomes available
- User receives overdue notices for unreturned items
- User can customize notification preferences (email, SMS, in-app)
- User receives alerts for new resources matching their interests

Manage Personal Profile

As a student or faculty member

I want to view and update my library account information

So that I can keep my contact details current and track my account status

Acceptance Criteria

- User can view their borrowing limits and current usage
- User can see outstanding fines and payment history
- User can update contact information and notification preferences
- User can view account restrictions or blocks
- System displays membership expiration date

Request New Resources

As a student or faculty member

I want to suggest new books or resources for library acquisition

So that I can help expand the collection with relevant materials

Acceptance Criteria

- User can submit resource requests with title, author, and justification

- User can track the status of their requests (pending, approved, rejected, acquired)
- Librarian receives notifications of new requests
- User receives updates when their requested resource is added to the catalog
- System prevents duplicate requests for the same resource

Manage User Accounts

As a librarian/administrator

I want to create, update, and manage user accounts

So that I can control access and maintain accurate user records

Acceptance Criteria

- Librarian can register new students and faculty members
- Librarian can update user information and borrowing privileges
- Librarian can suspend or block accounts for policy violations
- Librarian can adjust borrowing limits based on user type
- System maintains audit log of all account changes

Handle Lost or Damaged Items

As a librarian/administrator

I want to process claims for lost or damaged resources

So that I can maintain inventory accuracy and assess appropriate charges

Acceptance Criteria

- Librarian can mark items as lost or damaged in the system
- System automatically calculates replacement costs based on resource value
- Librarian can apply charges to user accounts
- Librarian can process damage assessments with varying severity levels
- System removes lost items from circulation and updates availability

Manage Reservations Queue

As a librarian/administrator

I want to view and manage all active reservations

So that I can ensure fair distribution and prioritize requests

Acceptance Criteria

- Librarian can view all reservations by resource or user
- Librarian can manually adjust queue positions if needed
- System highlights expired holds that need processing
- Librarian can cancel reservations with user notification
- Dashboard shows reservations requiring immediate attention

Configure System Settings

As a librarian/administrator

I want to configure library policies and system parameters

So that I can customize the system to institutional requirements

Acceptance Criteria

- Admin can set borrowing periods for different user types
- Admin can configure fine rates and grace periods
- Admin can set maximum renewal limits and reservation durations
- Admin can define notification schedules and templates
- Admin can manage resource categories and classification schemes

Bulk Operations Management

As a librarian/administrator

I want to perform bulk operations on multiple resources or users

So that I can manage large-scale updates efficiently

Acceptance Criteria

- Librarian can import multiple resources via CSV or Excel files
- Librarian can bulk update resource locations or categories

- Librarian can send notifications to multiple users simultaneously
- Librarian can generate bulk checkout for course reserves
- System validates bulk operations and reports errors

Manage Course Reserves

As a librarian/administrator

I want to create and manage course reserve collections

So that I can support faculty teaching requirements

Acceptance Criteria

- Librarian can designate specific resources as course reserves
- System applies shorter loan periods for reserve items
- Librarian can link reserves to specific courses and faculty
- Students can browse reserves by course or instructor
- System prioritizes access for enrolled students

Monitor System Activity

As a librarian/administrator

I want to view real-time system activity and user actions

So that I can ensure smooth operations and identify issues

Acceptance Criteria

- Dashboard displays active users and current transactions
- System shows real-time checkout and return activities
- Admin can view failed login attempts and security alerts
- System logs all administrative actions for audit purposes
- Librarian can monitor popular search terms and browse patterns

Send Announcements

As a librarian/administrator

I want to broadcast announcements to all users or specific groups

So that I can communicate library updates and policy changes

Acceptance Criteria

- Admin can create announcements with title, content, and priority
- Admin can target specific user groups (students, faculty, all)
- Announcements display on user dashboard upon login
- Admin can schedule announcements for future dates
- System tracks announcement read status

Browse Featured Collections

As a student or faculty member

I want to explore curated collections and recommendations

So that I can discover relevant resources I might have missed

Acceptance Criteria

- User can view featured collections on homepage
- Collections are organized by theme, subject, or trending topics
- User can browse "New Arrivals" and "Popular Resources"
- System displays personalized recommendations based on history
- User can save collections for later reference

Leave Resource Reviews

As a student or faculty member

I want to rate and review resources I've used

So that I can help other users make informed decisions

Acceptance Criteria

- User can rate resources on a 5-star scale
- User can write text reviews with their feedback
- Reviews display on resource detail pages
- User can only review resources they've borrowed or accessed
- System moderates reviews for inappropriate content

Export Citations

As a student or faculty member

I want to generate citations for resources in various formats

So that I can easily reference them in my academic work

Acceptance Criteria

- User can generate citations in APA, MLA, Chicago, and IEEE formats
- System includes all necessary bibliographic information
- User can copy citations to clipboard or export to reference managers
- Citations are available for both physical and digital resources
- System supports batch citation export for multiple resources

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